

Chapter 1

Introduction to Statistics

Learning Objectives

- Discuss the reasons for studying statistics as an engineer.
- Identify basic statistical concepts.
- Identify the levels of measurements
- Discuss the sampling procedures
- Understand data visualization using several graphical devices

1.0 Introduction to Statistics

Statistics is a way to get information from data. Statistics is a discipline which is concerned with:

- summarizing information to aid understanding,
- drawing conclusions from data,
- estimating the present or predicting the future, and
- designing experiments and other data collection.

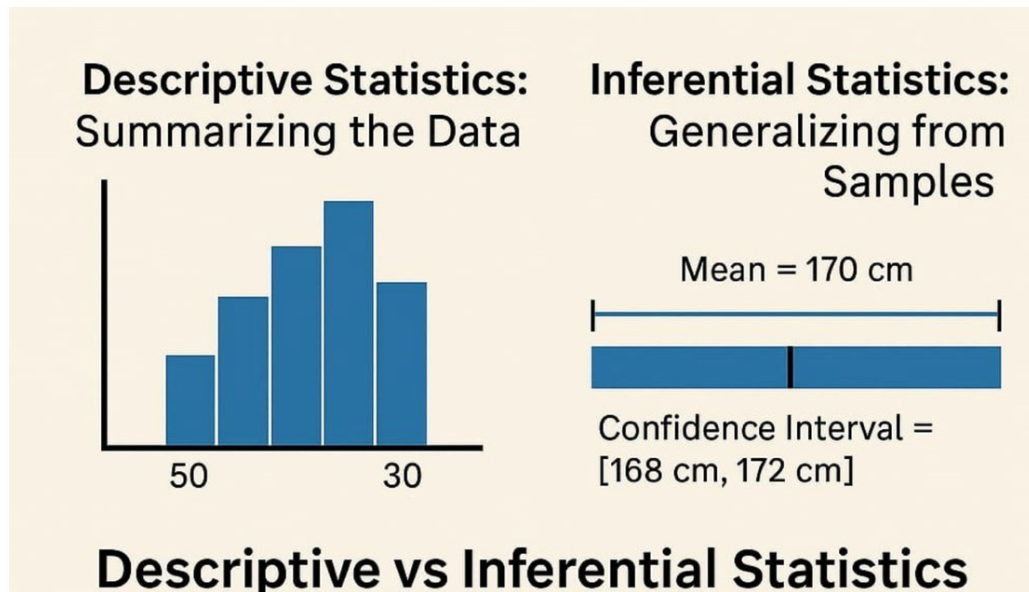
In making predictions, Statistics uses the concept of probability, which models chance mathematically and enables calculations of chance in complicated cases.

1.1 Why Statistics?

Engineering problems are rarely deterministic. Measurements vary, materials behave inconsistently, and environmental conditions change. Statistics equips engineers with the ability to analyze variability, assess reliability, and make decisions under uncertainty. Statistical methods are essential for designing new products, improving existing systems, optimizing processes, and evaluating performance. Through statistical analysis, engineers can distinguish meaningful signals from random noise, assess the quality of data, and quantify uncertainty in their conclusions. Beyond technical applications, statistical thinking supports objectivity in scientific inquiry. Conclusions drawn from data are strengthened when they are supported by formal statistical reasoning rather than subjective judgment. As a result, statistics plays a central role in research, quality control, risk assessment, and evidence-based decision-making across many disciplines. Even when engineers do not directly perform statistical analyses, understanding statistical concepts allows them to interpret results critically and make informed professional judgments.

1.2 Branches of Statistics

Statistics is traditionally divided into two interconnected branches: descriptive statistics and inferential statistics.



1.1.1 Descriptive Statistics

Descriptive statistics focuses on organizing, summarizing, and presenting data in a form that is easy to understand. This includes representing data visually through graphs and tables, as well as numerically through summary measures such as averages and measures of variability. The goal of descriptive statistics is to provide a clear and concise picture of the data at hand. At this stage, no generalizations are made beyond the observed dataset. Instead, descriptive statistics answers the question of what the data reveal about the observed phenomenon.

1.1.2 Inferential Statistics

Inferential statistics extends beyond the observed data by using information from a sample to draw conclusions about a larger population. Because it is often impractical or impossible to measure an entire population, inferential methods allow us to make informed statements about population characteristics based on limited data.

Probability theory underpins inferential statistics by providing a framework for quantifying uncertainty. Through estimation and hypothesis testing, inferential statistics enables researchers and engineers to assess the strength of evidence, evaluate competing explanations, and make predictions with known levels of confidence.

1.3 Variables and Types of Data

A variable is any characteristic or attribute that can take different values across individuals, objects, or observations. Variables form the basic building blocks of statistical analysis.

Variables can be broadly classified as qualitative or quantitative. Qualitative variables describe characteristics or categories and are typically expressed as labels or names. Examples include material type, defect category, or color classification. Quantitative variables, on the other hand, represent numerical measurements and express quantities such as weight, temperature, pressure, or time.

Quantitative variables can be further classified based on how they take values. Discrete variables arise from counting processes and take distinct, separate values, while continuous variables arise

from measurement processes and can take any value within a specified range. This distinction is important because it influences the choice of statistical methods and graphical representations.

	Quantitative Variables	Qualitative Variables
Definition	<i>Take on numeric values</i>	<i>Take on names or labels</i>
Examples	Number of students in a class	Eye color
	Number of square feet in a house	Gender
	Population size of a city	Breed of dog
	Age of an individual	Level of Education
	Height of an individual	Marital status

1.4 Univariate and Bivariate Data

Statistical data can also be classified based on the number of variables involved in the analysis. When data consist of measurements on a single variable, the data are referred to as univariate. Univariate analysis focuses on understanding the distribution, central tendency, and variability of that variable.

When two variables are measured simultaneously for each observation, the data are described as bivariate. Bivariate analysis explores relationships between variables, such as associations, trends, or dependencies. Understanding whether and how two variables are related is a key objective in many engineering and scientific studies.

1.5 Populations and Samples

In statistics, a population refers to the complete set of all individuals, objects, or measurements of interest in a study. A sample is a subset of the population selected for analysis. Because studying an entire population is often impractical, samples are used to represent populations.

The manner in which a sample is selected is critical. Well-designed sampling procedures aim to ensure that samples are representative of the population, thereby allowing valid inference. A commonly used approach is simple random sampling, in which every member of the population has an equal chance of being selected.

Numerical characteristics that describe a population are called parameters, while numerical characteristics computed from sample data are called statistics. For example, the population mean is denoted by the symbol μ , whereas the sample mean is denoted by x . One of the central goals of statistics is to use sample statistics to estimate unknown population parameters.



1.6 Graphical and Numerical Summaries of Data

Graphical displays play a crucial role in understanding data. For qualitative data, graphs such as bar charts and pie charts are commonly used to display frequencies and proportions. For quantitative data, graphical tools such as histograms, box plots, dot plots, and stem-and-leaf displays reveal important features of the distribution, including its center, spread, and shape.

In addition to graphical summaries, numerical summary statistics provide concise descriptions of data. Measures of location, such as the mean, median, and mode, describe where the data tend to cluster. Measures of variability, including the range, variance, and standard deviation, describe how spread out the data are around the center. Together, graphical and numerical summaries provide complementary perspectives on the same dataset.

1.7 Describing Data Patterns

When analyzing data, statisticians commonly describe distributions in terms of several key features. These include the center of the distribution, the degree of variability or spread, and the overall shape of the distribution. Shape is often discussed in terms of symmetry, skewness, and the presence of multiple peaks.

Modern data analysis emphasizes not just computing summary measures but also interpreting what they reveal about the underlying process that generated the data.

1.8 Unusual Features in Data

Real-world data often contain unusual features that merit special attention. Gaps in a distribution occur when certain ranges of values have no observations, potentially indicating structural constraints or data collection issues. Outliers are extreme values that differ markedly from the rest of the data and may arise from measurement errors, rare events, or meaningful deviations.

A commonly used rule for identifying outliers is based on the interquartile range. Observations that lie more than 1.5 times the interquartile range below the first quartile or above the third quartile are often classified as outliers. Identifying and interpreting unusual features is an important step in responsible data analysis.

1.9 Comparing Data Sets

Comparing two or more datasets is a common task in engineering and scientific studies. Graphical tools such as side-by-side box plots, comparative histograms, and grouped bar charts allow analysts to assess similarities and differences across datasets.

When comparing distributions, attention is typically given to differences in center, variability, shape, and the presence of unusual features. A systematic comparison helps ensure that conclusions are based on evidence rather than visual impressions alone.

1.5 Summarizing data graphically

Selected graphs for qualitative data

- Pie chart
- Bar Chart
- (Also frequency distribution)

Selected graphs for Numerical data

- Box plot
- Dot plot
- Stem-and-leaf
- Histogram

1.5 Summarizing Data Graphically

One of the first and most important steps in statistical analysis is visualizing the data. Graphical summaries allow us to explore data quickly, detect patterns, identify unusual features, and guide further analysis. Before computing numerical summaries such as means or standard deviations, statisticians almost always begin by examining appropriate graphs.

The choice of a graphical display depends fundamentally on the type of data being analyzed. In general, different graphical tools are used for qualitative (categorical) data and quantitative (numerical) data. Using an inappropriate graph can obscure important information or lead to misleading interpretations.

1.4.1 Graphical Displays for Qualitative (Categorical) Data

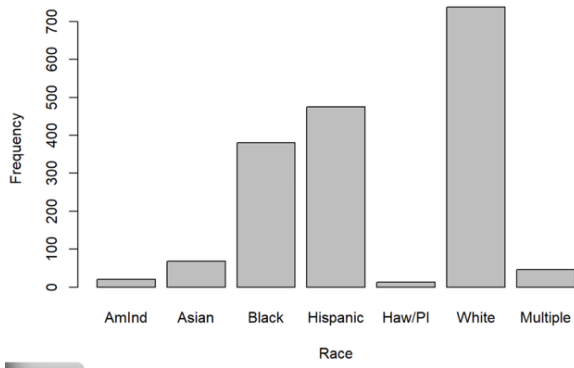
Qualitative data describe attributes or categories rather than numerical magnitudes. Graphs for qualitative data emphasize counts, proportions, and comparisons between categories.

The bar chart remains the most effective and widely used graphical tool for qualitative data. Each category is represented by a bar whose height corresponds to its frequency or relative frequency. Bar charts allow for easy comparison across categories and are particularly useful in engineering quality assessments, such as defect classification.

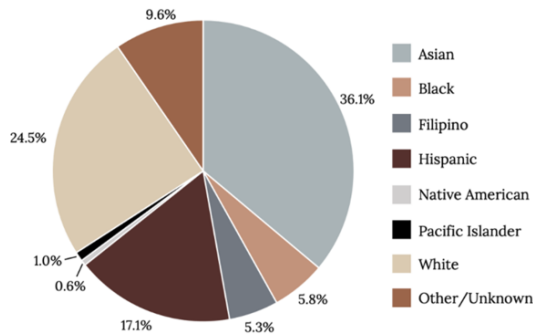
The pie chart displays each category as a sector of a circle, with sector sizes proportional to category percentages. Pie charts are useful for illustrating how categories contribute to a whole, although they are less effective when precise comparisons are required or when many categories are present.

A stacked bar chart extends the basic bar chart by showing subcategories within each bar. This type of graph is useful when comparing compositions across groups, such as defect types across multiple production lines.

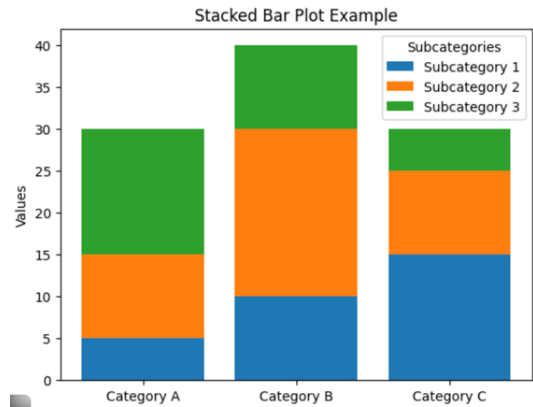
A Pareto chart is a specialized bar chart in which categories are ordered from most frequent to least frequent. Often used in quality control and industrial engineering, Pareto charts help identify the most significant contributors to a problem by visually reinforcing the “vital few versus trivial many” principle.

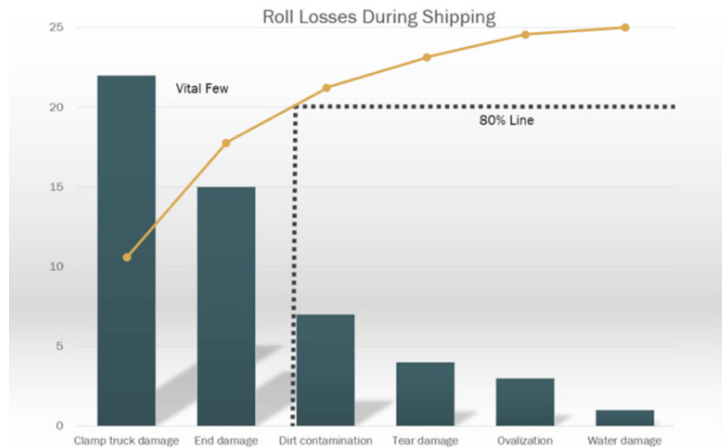


Ethnicity of Students



Pie chartPie chartBar chartBar chart



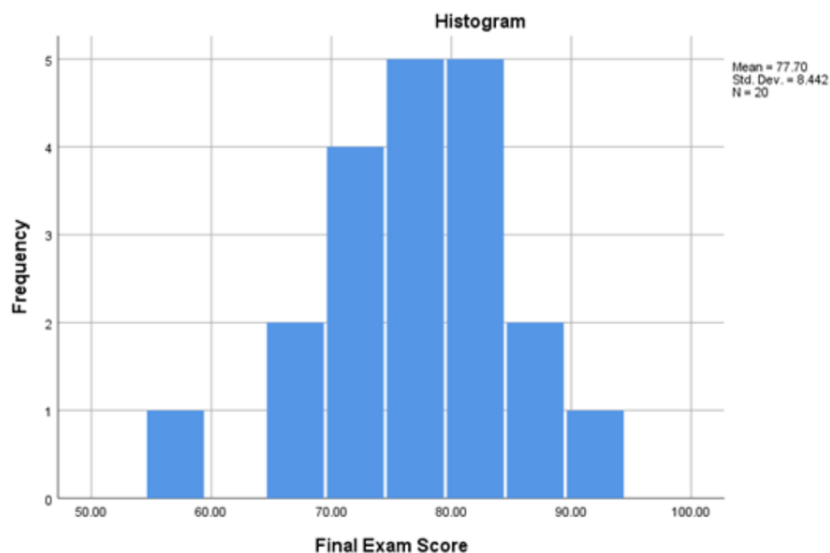


Pareto chart Pareto chart Stacked bar chart Stacked bar chart

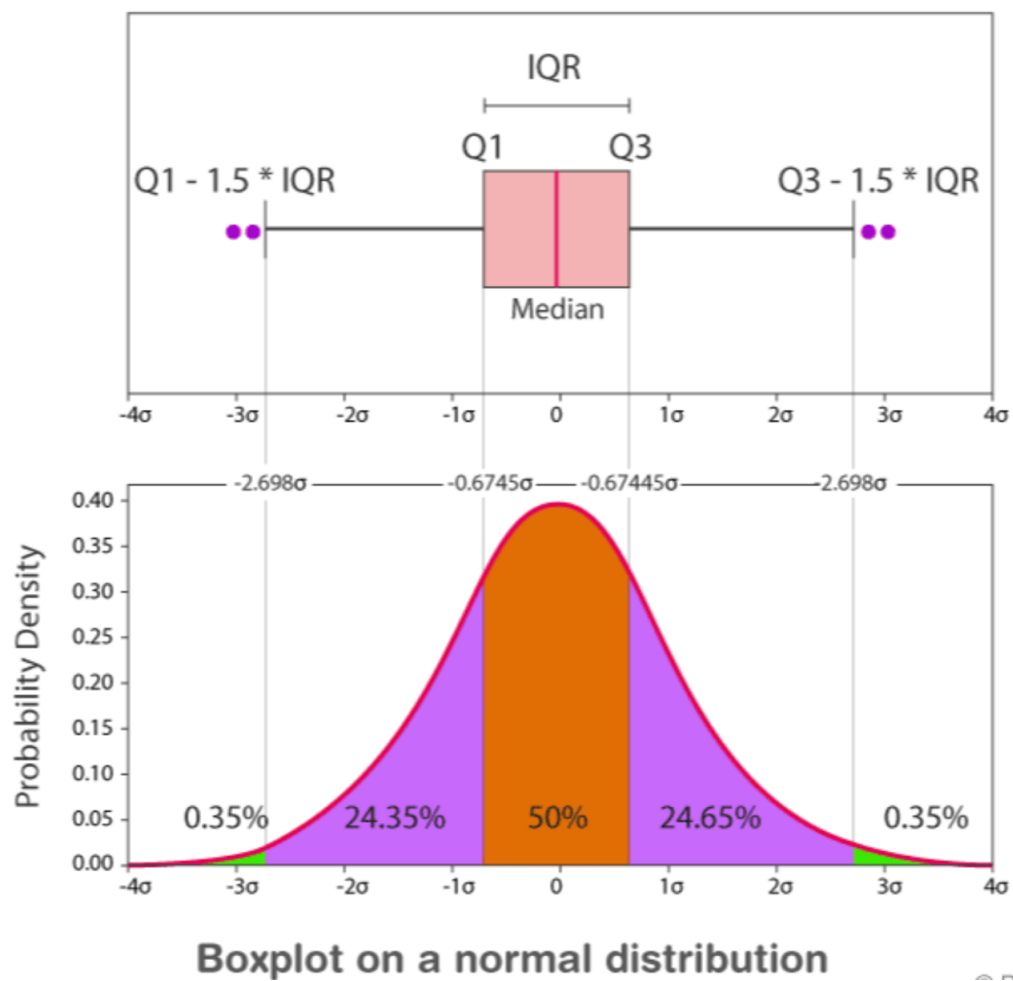
1.4.2 Graphical Displays for Quantitative (Numerical) Data

Quantitative data consist of numerical measurements, and their graphical summaries focus on revealing the distribution of values. Distributional graphs help answer questions about where data are centered, how spread out they are, and whether they exhibit skewness or extreme values.

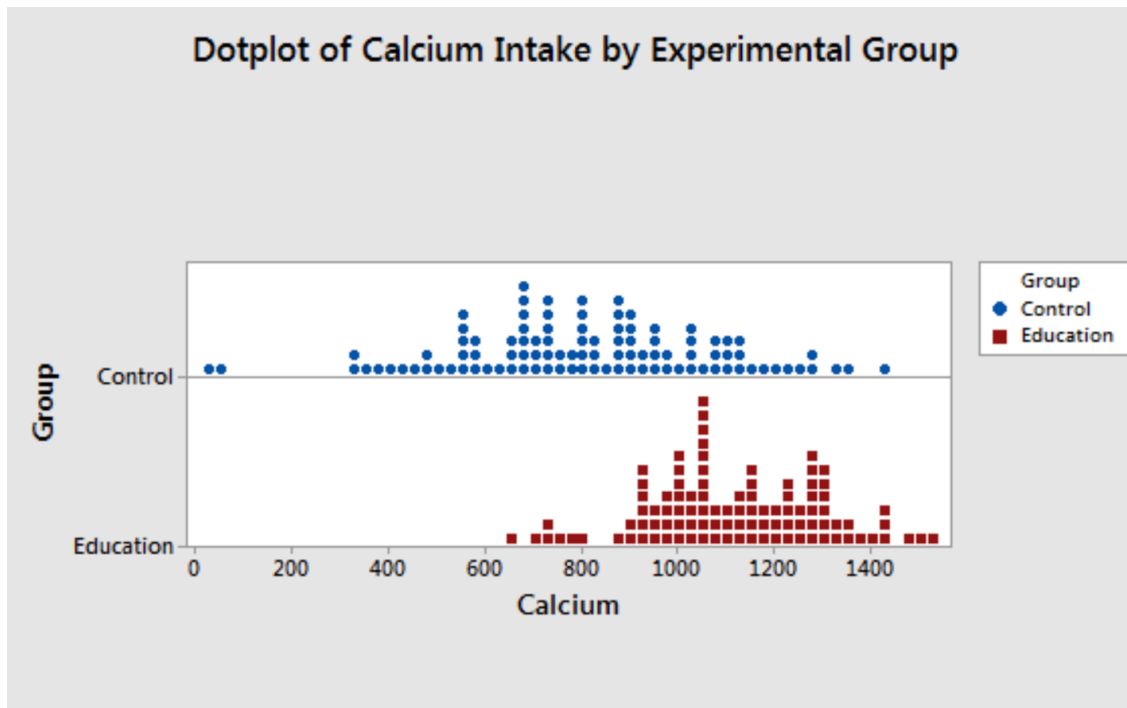
The histogram is one of the most important tools for exploring quantitative data. By grouping data into intervals, histograms show the overall shape of the distribution, including symmetry, skewness, and the presence of multiple modes. The choice of bin width can influence interpretation, making thoughtful construction essential.



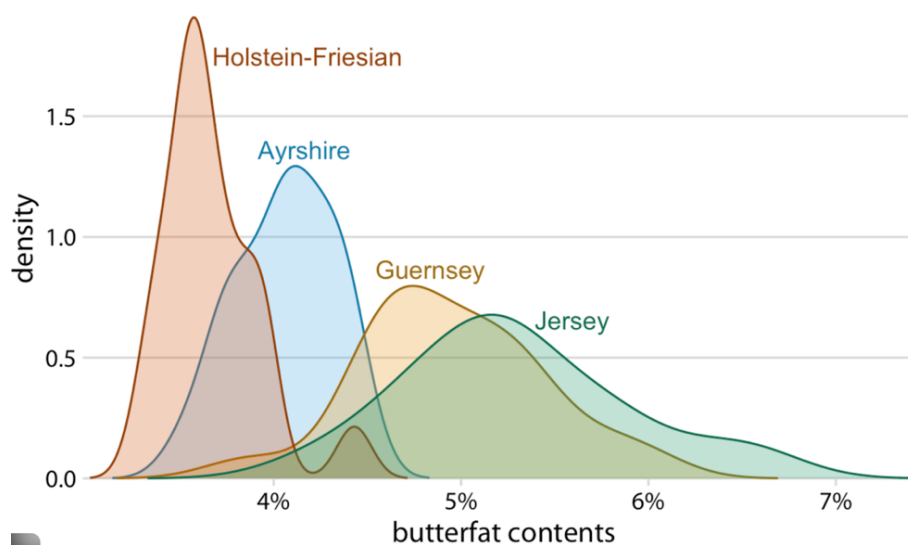
A box plot provides a concise summary of a distribution using the median, quartiles, and interquartile range. Box plots are especially valuable for comparing multiple datasets side by side and for identifying potential outliers.



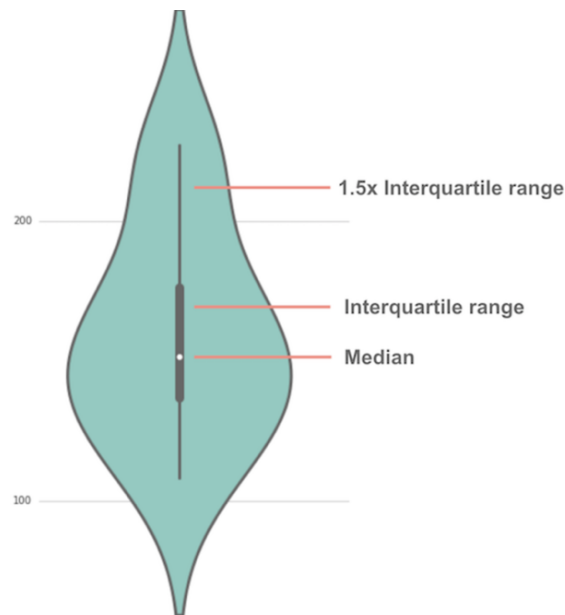
A dot plot displays individual observations along a number line. Because every data point is visible, dot plots are particularly useful for small to moderate datasets and for teaching purposes, as they preserve the raw data while still revealing patterns.



A density plot is a smoothed version of a histogram that estimates the probability density function of the data. Density plots are especially useful for comparing distributions and are widely used in modern statistical software.



A violin plot combines features of box plots and density plots. It displays the distribution's shape while also showing summary statistics. Violin plots are particularly effective when comparing distributions across groups.



Conceptual Takeaway for Students Students should understand that graphs are analytical tools, not decorations. A well-chosen graph can reveal insights that numerical summaries alone cannot. Developing the habit of visualizing data early in the analysis process is a defining characteristic of strong statistical reasoning and effective engineering practice.

1.6 Summary Statistics

While graphical displays allow us to visualize data, summary statistics provide concise numerical descriptions of key features of a dataset. Together, graphical and numerical summaries form the foundation of exploratory data analysis. Summary statistics are commonly grouped into measures of location and measures of variability.

Measures of Location (Central Tendency)

Measures of location describe where the data tend to cluster, or what might be considered a “typical” value in the dataset. The most commonly used measures of location are the mean, median, and mode.

The mean is the arithmetic average of the data and is widely used in engineering and scientific applications. However, it is sensitive to extreme values. The median is the middle value when the data are ordered and provides a measure of center that is resistant to outliers. The mode is the most frequently occurring value and is particularly useful for qualitative data or for identifying common outcomes.

In engineering practice, the choice of an appropriate measure of location depends on the nature of the data and the presence of unusual values. For example, median lifetimes are often preferred over means when analyzing component failure times due to skewness.

1.5.1 Measures of Spread (Variability)

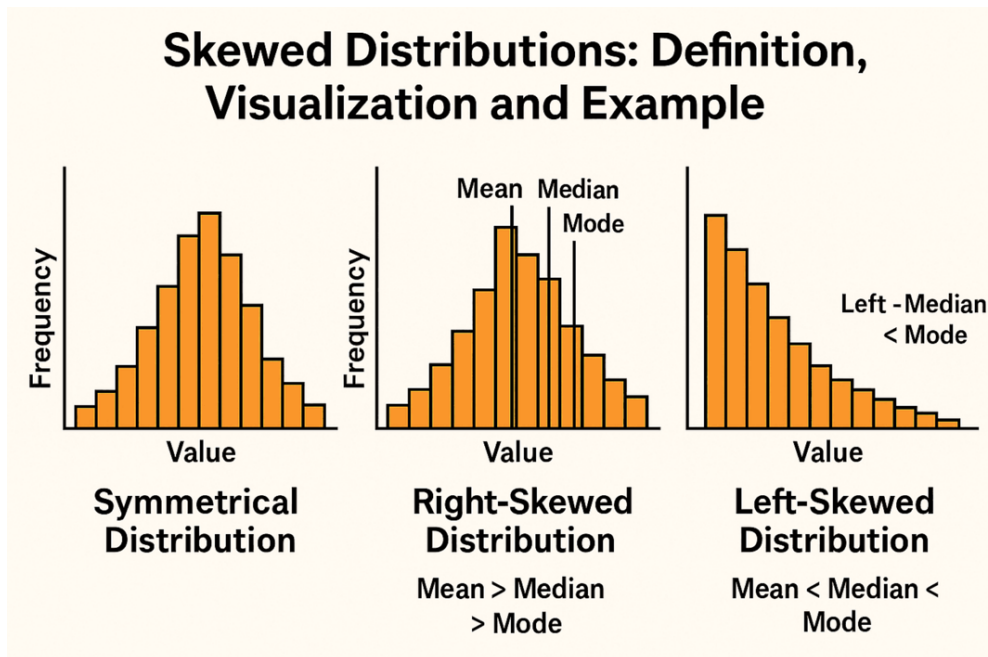
Measures of spread describe how dispersed the data are around the center. Two datasets may have the same mean but very different variability, leading to different engineering implications.

The range is the difference between the maximum and minimum values and provides a simple measure of spread. The variance measures the average squared deviation from the mean and forms the basis for many statistical methods. The standard deviation, which is the square root of the variance, is the most commonly used measure of variability because it is expressed in the same units as the original data.

In manufacturing and quality control, low variability is often as important as achieving a target mean, since excessive variation can lead to defects and system failure.

1.5.2 Describing Data Patterns

Beyond center and spread, statisticians describe data using additional features that capture the overall pattern of the distribution. These features are most easily recognized through graphical displays



The shape of a distribution refers to its general appearance. A distribution may be symmetric or skewed to the left or right. Skewness indicates the direction and degree of asymmetry, while kurtosis describes the heaviness of the tails and the peakedness of the distribution. Understanding these features helps engineers interpret the underlying processes generating the data. For example, right-skewed distributions are common in waiting times and failure data, while symmetric distributions often arise from measurement error around a stable process.

1.7 Unusual Features in Data

Real-world data rarely follow idealized patterns perfectly. As a result, statisticians pay close attention to unusual features that may indicate important phenomena, data collection issues, or rare but critical events.

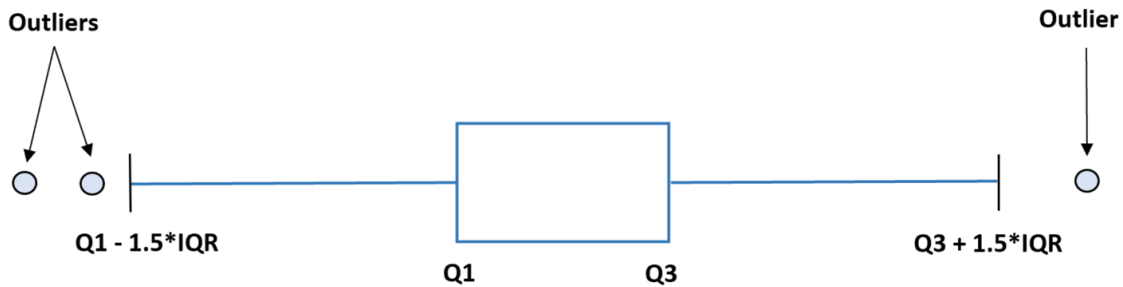
The two most common unusual features are gaps and outliers.

1.6.1 Gaps

A gap refers to a region within the range of the data where no observations occur. Gaps may arise due to physical constraints, changes in operating conditions, or data recording errors. In some cases, a gap may indicate the presence of distinct subgroups within the data, suggesting that a single distribution may not adequately describe the dataset.

1.6.2 Outliers

An outlier is an observation that lies unusually far from the rest of the data.



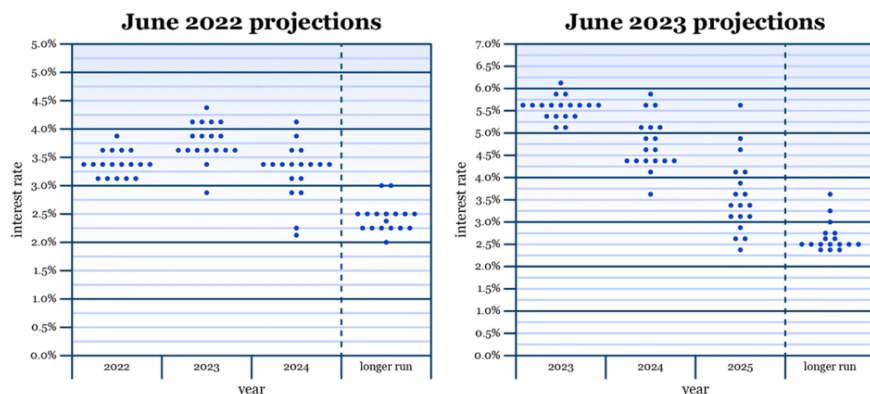
Outliers can result from measurement error, data entry mistakes, or genuinely extreme events. A commonly used rule for identifying outliers is based on the interquartile range (IQR). Observations lying more than 1.5 IQRs below the first quartile or above the third quartile are often flagged as potential outliers.

In engineering applications, outliers should never be removed automatically. Instead, they should be investigated carefully, as they may signal system faults, safety risks, or rare but important operating conditions.

1.8 Comparing Data Sets

In many practical situations, the goal is not to analyze a single dataset but to compare two or more datasets. Graphical displays are especially powerful for this purpose, as they allow direct visual comparison across groups or conditions.

Dot plot comparison: June 2022 vs. June 2023



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Source: federalreserve.gov

Common graphical tools for comparison include side-by-side box plots, comparative histograms,

dot plots, and bar charts. These displays help reveal differences and similarities that may not be apparent from numerical summaries alone.

1.7.1 Key Features for Comparing Data Sets

When comparing multiple datasets, statisticians focus on four fundamental features. The center indicates typical values, the spread reflects variability, the shape reveals distributional characteristics such as skewness or multimodality, and unusual features highlight gaps or outliers. A systematic comparison across these features ensures that conclusions are grounded in evidence rather than visual impression or single summary measures.

1.9 Sampling Procedures

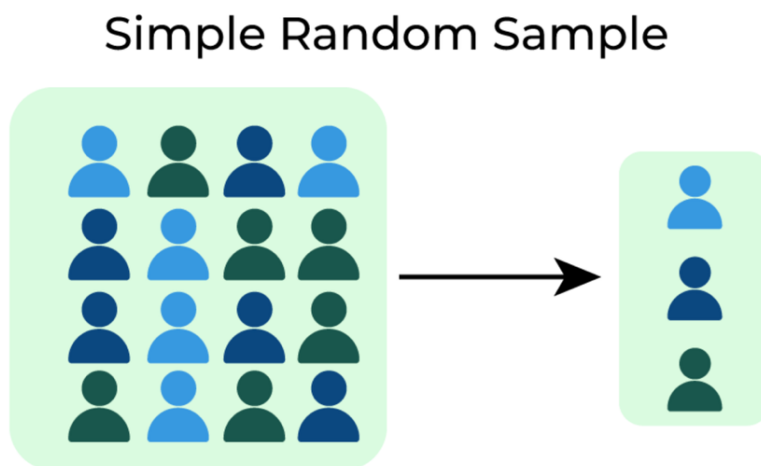
In statistical practice, it is rarely feasible to observe every element of a population. Instead, statisticians rely on sampling procedures to select a subset of observations, called a sample, from which conclusions about the population can be drawn. The fundamental goal of any sampling procedure is to obtain a sample that is representative of the population, meaning it accurately reflects the population's key characteristics.

A sampling procedure is said to be random if every possible sample of a given size has a known and non-zero probability of selection. Random sampling protects against systematic bias and provides the theoretical foundation for statistical inference. Without randomness, probability-based statements about uncertainty and confidence are no longer valid.

Different sampling procedures are used depending on the structure of the population, the objectives of the study, and practical constraints such as cost and accessibility.

1.8.1 Simple Random Sampling

Simple random sampling is the most fundamental sampling method and serves as the conceptual benchmark against which other methods are compared. In simple random sampling, every possible sample of size n drawn from a population of size N has equal probability of being selected.



For a finite population of size N , there are

$$\binom{N}{n}$$

distinct samples of size n . A sample is considered a simple random sample if each of these possible samples has probability $1/\binom{N}{n}$ of being chosen.

For an infinite population, or for populations that are very large relative to the sample size, a sample is considered random if each observation is selected independently from the others and according to the same probability distribution. In practice, many large populations can be treated as infinite when the sample size is small relative to the population size.

Example 1

There are 26 letters in the English alphabet. The total number of distinct samples of 5 letters that can be drawn is

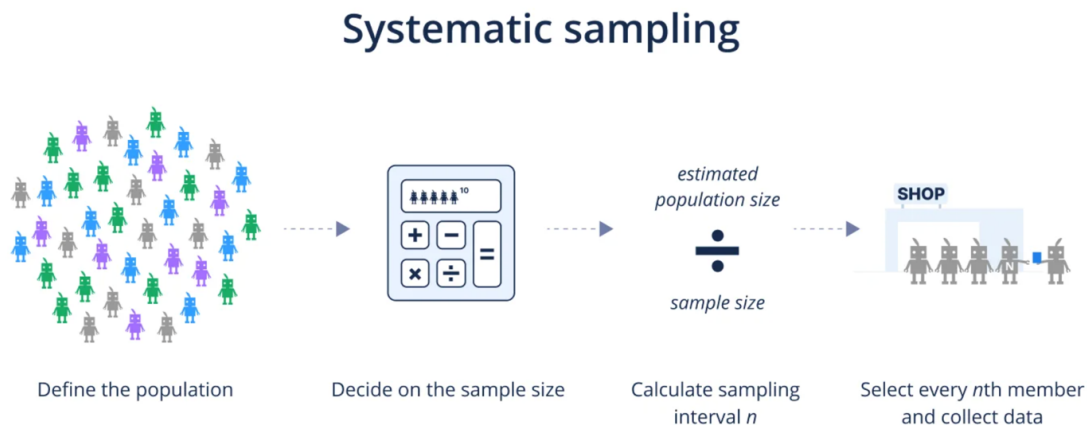
$$\binom{26}{5} = 65,780.$$

If a procedure is designed so that each of these 65,780 possible samples has an equal probability of being selected, then the resulting sample is a simple random sample. No letter or combination of letters is favoured over another.

Simple random sampling is conceptually ideal, but in practice it can be difficult to implement when populations are large, geographically dispersed, or poorly defined.

1.8.2 Systematic (Regular) Sampling

Systematic sampling is a structured alternative to simple random sampling that is often easier to implement in industrial and engineering settings. In this method, elements are selected at regular intervals from an ordered population.



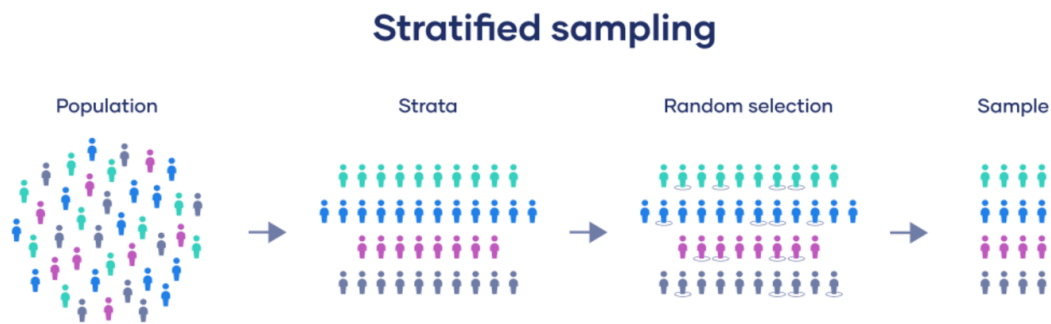
The procedure begins by randomly selecting a starting point from the first k elements of the population. Thereafter, every k -th element is selected for inclusion in the sample. The value of k , known as the sampling interval, is typically determined by dividing the population size by the desired sample size.

Systematic sampling has the advantage that the selected observations are spread evenly across the population, which often leads to samples that are highly representative. However, care must be taken to ensure that the population does not contain hidden periodic patterns that coincide with the sampling interval.

Example 2

In inspecting a batch of 1,000 pipes for defects, an engineer may choose to inspect every 10th pipe. The inspected pipes would then be the 10th, 20th, 30th, and so on, up to the 1,000th pipe.

This method is efficient and easy to implement. However, if every 10th pipe happens to be produced by a specific machine or process stage, the sample may be biased and fail to represent the entire batch accurately. Thus, systematic sampling requires careful consideration of the production process.

1.8.3 Stratified Random Sampling

Stratified random sampling is designed to improve representativeness when a population consists of distinct subgroups that differ in important ways. In this approach, the population is divided into mutually exclusive and collectively exhaustive sub-populations, called strata, and random samples are drawn independently from each stratum.

The strata are chosen so that observations within each stratum are relatively homogeneous with respect to the parameter of interest, while differences between strata are meaningful. This structure often leads to more precise estimates than simple random sampling.

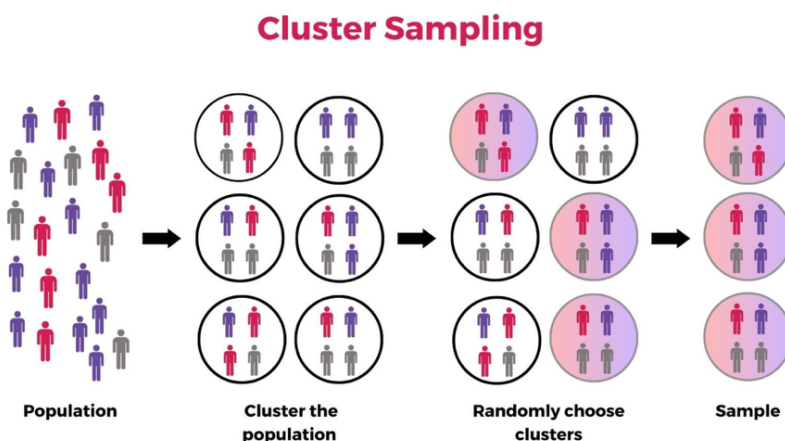
Example 3

To study income distribution among engineers in the Bay Area, the population can be divided into strata based on engineering discipline, such as electrical, mechanical, civil, chemical, and industrial engineering. Random samples are then selected from each discipline.

This approach is justified by the reasonable assumption that income levels are influenced by specialization. Stratified sampling ensures that all major engineering fields are adequately represented in the final sample.

1.8.4 Cluster Sampling

Cluster sampling is a sampling procedure in which the population is divided into groups, or clusters, and a random selection of these clusters is made. All elements within the selected clusters, or a random subset of them, are then included in the sample.



Unlike stratified sampling, clusters are typically chosen so that each cluster is a miniature representation of the entire population, rather than being internally homogeneous. Cluster sampling is particularly useful when populations are geographically dispersed or when obtaining a complete sampling frame is difficult.

Example 4

In conducting a statewide voter preference poll, researchers may randomly select congressional districts and survey all voters, or a random selection of voters, within those districts. When clusters are chosen carefully, the resulting sample can provide a reliable picture of statewide voter preferences at a fraction of the cost of simple random sampling.

1.10 Levels of Measurement (Types of Data)

Variables can be classified on the basis of their level of measurement. The way we classify variables greatly affects how we can use them in our analysis. Variables can be (1) nominal, (2) ordinal, (3) interval, or (4) ratio.

A nominal measurement is created when names are used to establish categories into which variables can be exclusively recorded. For example, sex can be classified as "male" or "female." You could also code it with a "1" or a "2"; oxide and carbonate ores may also be coded 1 and 2, but the numbers would serve only to indicate the categories and would carry no numerical significance; mathematical calculations using these codes would be meaningless. Soft drinks may be classified as Coke, Pepsi, 7-Up, or Ale 8. Each drink could be recorded in one of these categories to the exclusion of the others.

Nominal Measurements Names or classifications are used to divide data into separate and distinct categories.

It is important to remember that a nominal measurement carries no indication of order of preference, but merely establishes a categorical arrangement into which each observation can be placed.

Unlike a nominal measurement, an ordinal scale produces a distinct ordering or arrangement of the data. That is, the observations are ranked on the basis of some criterion. A retail company may rank its products as "good," "better," and "best." Opinion polls often use an ordinal scale such as "strongly agree," "agree," "no opinion," "disagree," and "strongly disagree."

As with nominal data, numbers can be used to order the rankings. Like nominal data, the magnitude of the numbers is not important; the ranking depends only on the order of the values. The retailer could have used the rankings of "1," "2," and "3," or "1," "3," and "12," for that matter. The arithmetic differences between the values are meaningless. A product ranked "2" is

not twice as good as one with a ranking of "1."

Ordinal Measurements Measurements that rank observations into categories with a meaningful order.

Variables on an interval scale are measured numerically, and, like ordinal data, carry an inherent ranking or ordering. However, unlike the ordinal rankings, the differences between the values is important. Thus, the arithmetic operations of addition and subtraction are meaningful. The Fahrenheit scale for temperatures is an example of an interval scale. Not only is 70 degrees hotter than 60 degrees, but also the same difference of 10 degrees exists as between 90 and 100 degrees Fahrenheit.

The value of zero is arbitrarily chosen in an interval scale. There is nothing sacrosanct about the temperature of zero; it is merely an arbitrary reference point. The Fahrenheit scale could have been created so that zero was set at a much warmer (or colder) temperature. No specific meaning is attached to zero other than to say it is 10 degrees colder than 10 degrees Fahrenheit. Thus, 80 degrees is not twice as hot as 40 degrees and the ratio $80/40$ has no meaning.

Interval Measurements Measurements on a numerical scale in which the value of zero is arbitrary but the difference between values is important.

Of all four levels of measurement, only the ratio scale is based on a numbering system in which zero is meaningful. Therefore, the arithmetic operations of multiplication and division also take on a rational interpretation. A ratio scale is used to measure many types of data found in business and geoscientific analyses. Variables such as costs, profits, inventory levels and grades are expressed as ratio measures. The value of zero dollars to measure revenues, for example, can be logically interpreted to mean that no sales have occurred. Furthermore, a firm with a 40 percent market share has twice as much of the market as a firm with a 20 percent market share. Measurements such as weight, time, and distance are also measured on a ratio scale since zero is meaningful, and an item that weighs 100 pounds is one-half as heavy as an item weighing 200 pounds.

Ratio Measurements Numerical measurements in which zero is a meaningful value and the difference between values is important.

You may notice that the four levels of measurement increase in sophistication, progressing from the crude nominal scale to the more refined ratio scale. Each measurement offers more information about the variable than did the previous one. This distinction among the various degrees of refinement is important, since different statistical techniques require different levels of measurements. While most statistical tests require interval or ratio measurements, other tests, called nonparametric tests (which will be examined later in this text), are designed to use nominal or ordinal data.

1.11 Measures of Location and Dispersion

In addition to graphical summaries such as histograms and frequency curves, the information contained in a dataset can be summarized numerically. Two numerical characteristics are of primary importance: the **location** of the data and the **dispersion** of the observations.

Measures of location, also called **measures of central tendency**, describe a value that is typical or representative of the dataset. They indicate where the observations tend to cluster. Measures of dispersion describe how widely the observations are spread around this central value. A meaningful statistical description of data must consider both aspects together.

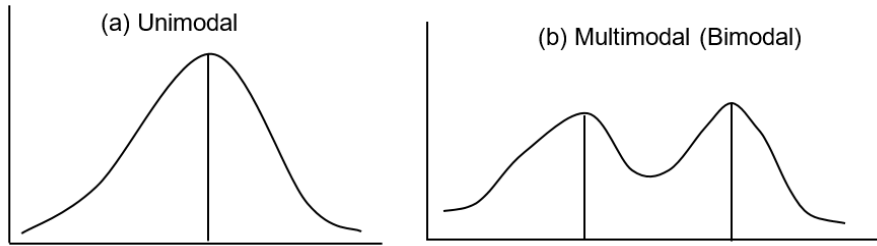
This section focuses on **measures of location**.

1.10.1 Measures of Location

Different measures of location capture different ideas of what is meant by a “typical” value. The most commonly used measures of location are the mode, the mean, and the median.

(a) The Mode

The mode is the value that occurs most frequently in a dataset. A dataset with a single mode is unimodal, while a dataset with multiple modes is multimodal. The mode is especially useful for qualitative data and for identifying the most common outcome in a set of observations.



Example 1

Consider the number of defects observed on identical manufactured components in a production batch:

2, 3, 1, 2, 0, 2, 1, 3, 2

The value 2 occurs most frequently. Therefore, the mode of the dataset is 2 defects per component. If two or more values occur with the same highest frequency, the dataset is multimodal, indicating that there is no single most typical value.

(b) The Mean

The mean is the most widely used measure of location. In general terms, the mean represents the “balance point” of the data. Several types of means are used in statistics, each appropriate for different types of data and applications.

(i) Arithmetic Mean (AM)

The arithmetic mean is defined as the sum of all observations divided by the number of observations. If a variable x takes values $x_1, x_2, \dots, x_n$, the arithmetic mean is denoted by $\bar{x}$.

The arithmetic mean uses all observations in the dataset, but it is sensitive to extreme values.

For a set of observations $x_1, x_2, \dots, x_n$, the arithmetic mean is defined as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The arithmetic mean is the most widely used measure of location and incorporates all observations. However, it is sensitive to extreme values.

Example 2

Suppose the measured lengths (in mm) of five machined parts are:

98, 100, 101, 99, 102

The arithmetic mean length is:

$$\bar{x} = \frac{98 + 100 + 101 + 99 + 102}{5} = 100\text{mm}$$

This value represents the average length of the parts. For large datasets, data are often grouped

into classes. In such cases, the arithmetic mean is computed using class marks and frequencies, which provides an efficient approximation for large volumes of data.

(ii) Geometric Mean

The geometric mean is defined only for positive, non-zero values. It is obtained by taking the n th root of the product of the observations. The geometric mean is particularly appropriate when dealing with percentage changes, growth rates, and multiplicative processes. The geometric mean answers the question:

“What constant growth rate would produce the same overall change?”

The geometric mean is defined for positive, non-zero values and is given by:

$G = (\prod_{i=1}^n x_i)^{1/n}$ The geometric mean is appropriate for multiplicative processes, such as growth rates and percentage changes.

Example 3 (Growth rate application)

Suppose the annual growth rates of a company's output over three years are:

+10%, +15%, and +5%

Expressed as growth factors, these are:

1.10, 1.15, and 1.05

The geometric mean growth factor is:

$$G = \sqrt[3]{1.10 \times 1.15 \times 1.05}$$

This value represents the average annual growth rate over the three years. Because growth compounds multiplicatively, the geometric mean provides a more meaningful summary than the arithmetic mean in such situations. It is always true that the geometric mean is less than or equal to the arithmetic mean, with equality only when all observations are equal.

(iii) Harmonic Mean

The harmonic mean is primarily used to compute average rates, particularly when a fixed quantity, such as distance or workload, is involved. It is defined as the reciprocal of the arithmetic mean of the reciprocals of the observations. The harmonic mean is appropriate when averaging quantities expressed as “per unit,” such as speed or efficiency. The harmonic mean is used to compute average rates when a fixed quantity such as distance or workload is involved.

It is defined as:

$$H = \frac{n}{\sum \frac{1}{x_i}}$$

Example 4

An engineer drives a test vehicle over the same distance at two different speeds:

60 km/h on the first segment and 40 km/h on the second segment.

The average speed over the entire trip is not the arithmetic mean of 60 and 40. Instead, it is given by the harmonic mean:

$$H = \frac{2}{\frac{1}{60} + \frac{1}{40}} = 48 \text{ km/h}$$

This example illustrates why the harmonic mean is the correct measure when distances are fixed. For any set of positive numbers, the following inequality always holds:

$$\text{Harmonic Mean} \leq \text{Geometric Mean} \leq \text{Arithmetic Mean}$$

(c) The Median

The median is the middle value of an ordered dataset. If the number of observations is odd, the median is the central value. If the number of observations is even, the median is the average of the two central values. Unlike the mean, the median is resistant to extreme values, making it particularly useful for skewed data.

Median for Ungrouped Data

Let $x_1, x_2, \dots, x_n$ denote a random sample of size n , arranged in increasing order of magnitude.

If n is odd, the median is the observation in position

$$\frac{n+1}{2}$$

If n is even, the median is the average of the observations in positions

$$\frac{n}{2} \text{ and } \frac{n}{2} + 1$$

That is,

$$\text{Median} = \begin{cases} x_{\frac{n+1}{2}}, & \text{if } n \text{ is odd} \\ \frac{1}{2} \left(x_{\frac{n}{2}} + x_{\frac{n}{2}+1} \right), & \text{if } n \text{ is even} \end{cases}$$

Median for Grouped Data

When data are grouped into class intervals, the exact median cannot be determined directly. Instead, it is estimated using the median class, which is the class interval containing the median position.

The general formula for the median of grouped data is:

$$\text{Median} = b_L + \left(\frac{\frac{n}{2} - f_{m-1}}{f_m} c \right)$$

where:

- b_L = lower boundary of the median class
- n = total number of observations
- f_{m-1} = cumulative frequency of the class preceding the median class
- f_m = frequency of the median class
- c = class width

This formula assumes that observations are uniformly distributed within the median class.

Example 5 (Ungrouped data)

Consider the annual salaries (in thousands) of five engineers:

45, 48, 50, 52, 120

The arithmetic mean is inflated by the extremely high salary, while the median (50) better represents the typical salary in the group.

Example 6 (Grouped data)

Given:

- $b_L = 199.5$
- $n = 300$
- $f_{m-1} = 116$
- $f_m = 73$
- $c = 50$

Substituting into the formula:

$$\text{Median} = 199.5 + \left(\frac{150 - 116}{73} \right) 50$$

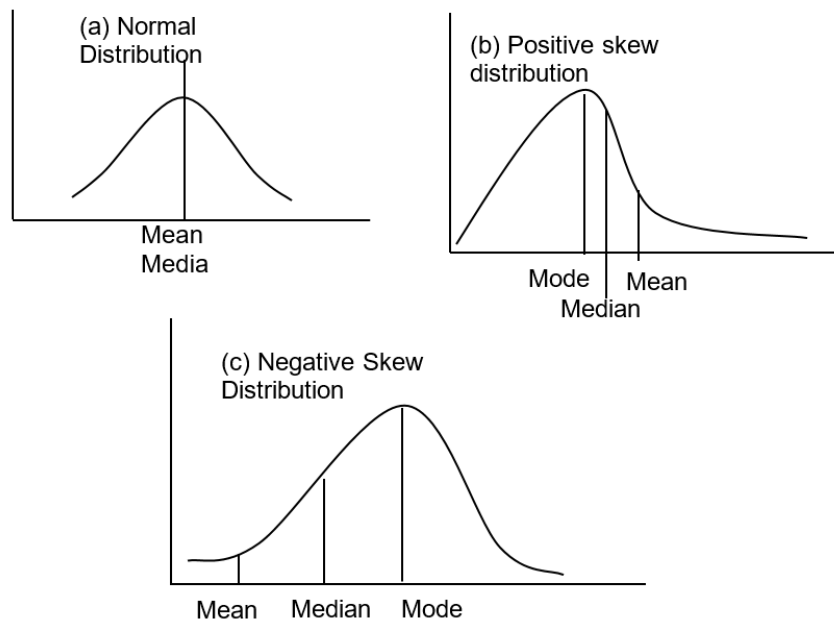
$$\text{Median} = 199.5 + (0.4658)(50)$$

$$\text{Median} \approx 222.8$$

(d) Relationship Between Measures of Location and Distribution Shape

The relative positions of the mode, median, and mean depend on the shape of the frequency distribution. In symmetrical distributions, all three measures coincide. In positively skewed distributions, the mean is greater than the median, which is greater than the mode. In negatively skewed distributions, the mean is less than the median, which is less than the mode.

Understanding this relationship is essential for selecting an appropriate measure of location and for interpreting data correctly in engineering applications.



1.11 Measure of Location and Dispersion

In addition to the histogram, the information in the frequency distribution can be further summarised by means of just two numbers. The first is the location of the data, and the various numbers that provide information about this are known as 'measures of location' or 'measures of central tendency'. 'Location of the data' refers to a value that is typical of all the sample observations.

The second important aspect of the data is the dispersion of the observations. This implies how the data are scattered (dispersed). This is also called 'measure of variation'.

1.12 Measures of Dispersion

Measures of location describe the central or typical value of a dataset, but they do not reveal how the data are distributed around that central value. Two datasets may have the same mean or median yet differ greatly in their degree of variability. Measures of dispersion quantify this variability and describe how widely the observations are spread.

In engineering and scientific applications, dispersion is critically important. A process may achieve the correct average output, but if the variability is too high, the process may still be unreliable or unsafe. Measures of dispersion therefore provide insight into consistency, precision, stability, and risk.

Measures of dispersion can be broadly classified into three categories:

1. Measures based on extreme or selected values,
2. Measures based on deviations from a central value,
3. Measures based on squared deviations.

1.11.1 Measures Based on Extreme or Selected Values

(a) The Range

The range is the simplest measure of dispersion and is defined as the difference between the largest and smallest observations in the dataset:

$$\text{Range} = x_{\max} - x_{\min}$$

Example 1

The tensile strengths (MPa) of five steel samples are:
410, 415, 420, 418, 412

$$\text{Range} = 420 - 410 = 10 \text{ MPa}$$

The range gives a quick indication of spread but depends only on two values and ignores the behavior of the remaining observations.

(b) The Interquartile Range (IQR)

The interquartile range (IQR) measures the spread of the central 50% of the data and is defined as:

$$\text{IQR} = Q_3 - Q_1$$

where Q_1 and Q_3 are the first and third quartiles, respectively.

Example 2

If the 25th percentile of a dataset is 52 and the 75th percentile is 68:

$$\text{IQR} = 68 - 52 = 16$$

The IQR is resistant to extreme values and is commonly used in box plots and outlier detection.

(c) The Interdecile Range

The interdecile range measures the spread of the central 80% of the data and is defined as:

$$InterdecileRange = D_9 - D_1$$

This measure is mainly used in descriptive statistics because it is less convenient for theoretical analysis.

1.11.2 Measures Based on Deviations from a Central Value

(a) Mean Deviation

The mean deviation is defined as the average of the deviations of observations from a central value (usually the mean or median)

$$MeanDeviation = \frac{1}{n} \sum (x_i - x)$$

Because positive and negative deviations cancel out, this measure can be misleading and is rarely used in practice.

(b) Mean Absolute Deviation (MAD)

To avoid cancellation of deviations, the mean absolute deviation (MAD) uses absolute values:

$$MAD = \frac{1}{n} \sum |x_i - x|$$

Example 3

Consider the dataset:

6, 9, 10, 13, 17, 19, 21, 25

The absolute deviations from the mean are calculated and averaged to obtain the MAD.

The MAD is intuitive and uses all observations, but it is difficult to manipulate mathematically in advanced statistical analysis.

1.11.3 Variance and Standard Deviation

To overcome the limitations of absolute deviations, statisticians square the deviations from the mean. This leads to the variance, which is the foundation of many statistical methods.

(a) Variance

For a sample, the variance is defined as:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

The denominator $n - 1$ is used to obtain an unbiased estimate of the population variance.

For a population, the variance is:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

(b) Standard Deviation

The standard deviation is the positive square root of the variance:

$$s = \sqrt{s^2}$$

The standard deviation is expressed in the same units as the original data, making it the most interpretable measure of dispersion.

Types of Standard Deviation

(i) Population Standard Deviation

Used when data include every member of the population:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

(ii) Sample Standard Deviation

Used when data represent a sample from a larger population:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

This is the most common form used in engineering and scientific analysis.

(iii) Standard Deviation for Grouped Data

When data are grouped into classes, the standard deviation is computed using class midpoints:

$$s = \sqrt{\frac{\sum f_i (x'_i - \bar{x})^2}{n - 1}}$$

where:

- x'_i is the class midpoint,
- f_i is the class frequency,
- n is the total number of observations.

Example 4 (Sample Standard Deviation)

Consider the measured voltages (V):

98, 100, 102, 100, 100

Mean:

$$\bar{x} = 100$$

Squared deviations:

$$(98 - 100)^2 = 4, (100 - 100)^2 = 0, (102 - 100)^2 = 4$$

$$s^2 = \frac{4 + 0 + 4 + 0 + 0}{4} = 2$$

$$s = \sqrt{2} \approx 1.41V$$

This indicates that the voltages typically deviate from the mean by about 1.4 V.

Interpretation of Standard Deviation

- Small standard deviation → data tightly clustered → high precision
- Large standard deviation → data widely spread → low consistency

In quality control and reliability engineering, reducing standard deviation is often more important than shifting the mean.

1.11.4 Why Dispersion Matters in Engineering

Dispersion affects:

- product quality,
- system reliability,
- safety margins,
- risk of failure.

Two processes with identical means may perform very differently if their variability differs. Engineers must therefore evaluate dispersion before making decisions based on averages.

Key Insight for Students Measures of dispersion quantify uncertainty and reliability. In engineering practice, controlling variability is often more important than controlling the average. Key Insight for Students Measures of dispersion quantify uncertainty and reliability. In engineering practice, controlling variability is often more important than controlling the average.

1.14 Measures of Dispersion, Skewness

It should be clear that a measure of central tendency by itself can exhibit only one of the

PRACTISE QUESTIONS

Question 1

Two manufacturing machines produce steel rods intended to have a length of 100 mm.

Machine A produces rods with lengths (mm):

98, 99, 100, 101, 102

Machine B produces rods with lengths (mm):

90, 95, 100, 105, 110

- (i) Compute the arithmetic mean for both machines.
- (ii) Compute the range and standard deviation for both machines.
- (iii) Which machine would you recommend for mass production?
- (iv) Which measure would you report to a non-technical manager, and why?

Question 2

Test scores from an engineering statistics course are grouped into classes and summarized as follows:

Class Interval	Frequency
50–59	5
60–69	8
70–79	14
80–89	18
90–99	6

- (i) Estimate the mean and standard deviation using grouped-data formulas.

- (ii) Explain why the grouped-data standard deviation is only an approximation.
- (iii) Under what circumstances is grouping unavoidable in engineering practice?

Question 3

A dataset describing waiting times for system failure is strongly right-skewed.

- (i) Would the standard deviation still be an appropriate measure of dispersion?
- (ii) Which combination of location and dispersion measures would you recommend?
- (iii) Explain how skewness affects risk assessment.

THEORITICAL QUESTIONS

1. Explain the difference between descriptive statistics and inferential statistics. Give one engineering example where each is used.
2. Why is probability essential in inferential statistics? Explain in the context of making decisions from sample data.
3. Distinguish clearly between:
 - a population and a sample
 - a parameter and a statistic

Provide one example of each from engineering practice.

4. Explain why a sample can never perfectly represent a population, even when random sampling is used.
5. Why is it dangerous to rely on a single numerical summary (such as the mean) when describing a dataset?
6. Explain why a histogram is more appropriate than a bar chart for numerical data.
7. Under what circumstances would a box plot be preferred over a histogram?
8. Describe one situation where a Pareto chart would be more informative than a pie chart.

PROJECT ASSIGNMENT (100 %)

Project Purpose

This project is designed to help students apply the concepts learned in Chapter 1: Introduction to Statistics to a real-world situation. The emphasis is on data collection, descriptive analysis, and statistical interpretation, rather than on probability or inference. By completing this project, students will demonstrate their ability to:

- collect and organize data,
- summarize data graphically and numerically,
- assess variability and distribution shape,
- communicate statistical findings clearly and logically.

Project Overview

Each student (or approved group) must identify a real-world variable, collect original data, and conduct a complete descriptive statistical analysis using only the methods covered in Chapter 1. No inferential methods (confidence intervals, hypothesis tests, probability models) should be used.

Step 1: Selection of Variable and Context

Students must select multiple primary variables that is measurable and relevant to engineering, science, or everyday systems.

The chosen variables must be:

- quantitative,
- clearly defined,
- measurable using a consistent method.

Examples of acceptable variables include:

- dimensions of manufactured items,
- waiting or processing times,
- energy or water consumption,
- environmental measurements (temperature, noise level, illumination),
- test scores or laboratory results,
- number of defects per unit.

Students must clearly state:

- what is being measured,
- the unit of measurement,
- the context in which the data arise.

Step 2: Data Collection Plan

Before collecting data, students must describe their data collection procedure, including:

- where the data will be collected,
- how measurements will be taken,
- the tool or method used for measurement,
- how consistency will be ensured.

Data Requirements

- Minimum of 30 observations,
- Data must be collected personally or obtained from a clearly identified reliable source,
- All observations must relate to the same variable and context.

Any potential sources of measurement error or bias should be briefly discussed.

Step 3: Data Organization and Classification

Students should present the raw data in a clearly labeled table and arrange the data in ascending order. The variable should be classified as discrete or continuous, and the level of measurement (nominal, ordinal, interval, or ratio) should be identified. If appropriate, the data may be grouped into class intervals, with justification provided for the chosen class width.

Step 4: Graphical Representation

Students must construct at least three appropriate graphical displays selected from those studied in Chapter 1. These may include histograms, box plots, stem-and-leaf plots, dot plots, or bar or Pareto charts where appropriate. Each graph must include a clear title, labeled axes with units, and a short written interpretation describing the observed patterns.

Step 5: Measures of Location

Students must compute and interpret the mode, arithmetic mean, and median. Where appropriate, the geometric mean or harmonic mean may also be computed. For each measure, students should explain what the value represents in context, whether it is appropriate as a typical value, and how the shape of the distribution affects its interpretation.

Step 6: Measures of Dispersion

Students must compute and interpret the range, interquartile range, variance, and standard deviation. Where relevant, the coefficient of variation should also be computed. Interpretation should focus on the degree of variability, the consistency of the data, and the practical implications of high or low dispersion.

Step 7: Distribution Shape and Unusual Features

Students must analyze the shape of the distribution by discussing symmetry or skewness, the relative positions of the mean, median, and mode, and the presence of gaps or outliers. Kurtosis should be discussed qualitatively in terms of peakedness or tail behavior. The relationship between graphical features and numerical summaries should be clearly explained.

Step 8: Discussion and Interpretation

A discussion section should address the main patterns observed in the data, possible reasons for variability, limitations of the data collection process, and practical implications of the findings. The emphasis should be on interpretation rather than computation.

Step 9: Report Structure

The project report should be organized as follows:

Title and objective

Description of variable and data collection method

Data table

Graphical summaries

Measures of location

Measures of dispersion

Distribution shape analysis

Discussion and conclusions

Reports must be clearly written, logically organized, and neatly presented.

NOTE!!! Academic Integrity

All data must be original or properly referenced. Fabricated or copied datasets will not be accepted.